

## Initial Project Planning Template

|               |                                                                |
|---------------|----------------------------------------------------------------|
| Date          | 21 June 2026                                                   |
| Team ID       |                                                                |
| Project Name  | Rising Waters: A Machine Learning Approach to Flood Prediction |
| Maximum Marks | 5 Marks                                                        |

### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

| Sprint   | Functional Requirement (Epic)    | User Story Number | User Story / Task                                                                                                                                                             | Story Points | Priority | Team Members       | Sprint Start Date | Sprint End Date (Planned) |
|----------|----------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|--------------------|-------------------|---------------------------|
| Sprint-1 | User Interface & Authentication  | USN-1             | As a user, I can access a dashboard home page so that I can see project introduction scopes.                                                                                  | 2            | High     | Frontend Developer | 19 June 2026      | 21 June 2026              |
| Sprint-1 | User Interface & Authentication  | USN-2             | As a user, I can register and log into the application using secure email/password sessions.                                                                                  | 2            | Medium   | Frontend Developer | 19 June 2026      | 21 June 2026              |
| Sprint-1 | User Interface & Authentication  | USN-3             | As a user, I can access an input web form layout so that I can enter local weather and rainfall numbers.                                                                      | 2            | High     | Frontend Developer | 19 June 2026      | 21 June 2026              |
| Sprint-2 | Data Engineering & Visualization | USN-4             | As a user, I can register for the application through Gmail As a developer, I can ingest historical environmental tables using Kaggle APIs to set up data collection scripts. | 2            | High     | Data Engineer      | 22 June 2026      | 24 June 2026              |
| Sprint-2 | Data Engineering & Visualization | USN-5             | As a data analyst, I can plot distribution graphs, box charts, and correlation heatmaps to explore features.                                                                  | 3            | High     | Data Engineer      | 22 June 2026      | 24 June 2026              |

|          |                                  |        |                                                                                                                                        |   |        |                     |              |              |
|----------|----------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------|---|--------|---------------------|--------------|--------------|
| Sprint-2 | Data Engineering & Visualization | USN-6  | As a developer, I can handle missing values, isolate outliers, and apply features scaling variables.                                   | 3 | High   | Data Engineer       | 22 June 2026 | 24 June 2026 |
| Sprint-3 | Machine Learning Modeling        | USN-7  | As an ML engineer, I can train the XGBoost classifier to obtain high classification accuracy limits.                                   | 5 | High   | ML Engineer         | 25 June 2026 | 26 June 2026 |
| Sprint-3 | Machine Learning Modeling        | USN-8  | As an ML engineer, I can export the verified trained components to permanent <code>model.pkl</code> and <code>scaler.pkl</code> files. | 2 | High   | ML Engineer         | 25 June 2026 | 26 June 2026 |
| Sprint-4 | Backend Integration              | USN-9  | As a developer, I can build Flask route mappings to validate web form data before processing arrays.                                   | 3 | High   | Backend Developer   | 27 June 2026 | 28 June 2026 |
| Sprint-4 | Backend Integration              | USN-10 | As a developer, I can load the saved joblib serialization blocks into Flask memory layers to perform runtime calculations.             | 3 | High   | Backend Developer   | 27 June 2026 | 28 June 2026 |
| Sprint-4 | Backend Integration              | USN-11 | As a user, I can view distinct dynamic output landing screens displaying explicit "Flood Chance" or "No Flood Chance" results.         | 2 | High   | Backend Developer   | 27 June 2026 | 28 June 2026 |
| Sprint-5 | Cloud Deployment & Logging       | USN-12 | As a user, I can check a prediction history panel to review my account's historical search activities.                                 | 2 | Medium | Full-Stack / DevOps | 29 June 2026 | 30 June 2026 |
| Sprint-5 | Cloud Deployment & Logging       | USN-13 | As a developer, I can push local files to GitHub and set up server hosting setups onto the <b>IBM Cloud</b> pipeline.                  | 3 | High   | Full-Stack / DevOps | 29 June 2026 | 30 June 2026 |